


1.0.1 UGC NET CSE | June 2012 | Part 2 | Question: 2


The post order traversal of a binary tree is DEBFCA. Find out the pre-order traversal

- A. ABFCDE B. ADBFEC C. ABDECF D. None of the above

ugcnetcse-june2012-paper2 algorithms

Answer key

1.0.2 UGC NET CSE | August 2016 | Part 3 | Question: 36


Match the following :

- | | |
|-----------------------------|-------------------|
| a. Prim's algorithm | i. $O(V^2 E)$ |
| b. Bellman-Ford algorithm | ii. $O(V \lg V)$ |
| c. Floyd-Warshall algorithm | iii. $O(E \lg V)$ |
| d. Johnson's algorithm | iv. $O(V^3)$ |

where V is the set of nodes and E is the set of edges in the graph.

Codes :

- A. a-i, b-iii, c-iv, d-ii
 B. a-i, b-iii, c-ii, d-iv
 C. a-iii, b-i, c-iv, d-ii
 D. a-iii, b-i, c-ii, d-iv

ugcnetcse-aug2016-paper3 algorithms

Answer key

1.0.3 UGC NET CSE | July 2018 | Part 2 | Question: 30


Consider a Boolean function of 'n' variables. The order of an algorithm that determines whether the Boolean function produces a output 1 is

- A. Logarithmic B. Linear C. Quadratic D. Exponential

ugcnetcse-july2018-paper2 algorithms

Answer key

1.0.4 UGC NET CSE | July 2018 | Part 2 | Question: 27


Match the following with respect to algorithm paradigms :

List-I

- (a) The 8-Queen's problem
 (b) Single-Source shortest paths
 (c) STRASSEN's Matrix multiplication
 (d) Optimal Binary search trees

List-II

- (i) Dynamic programming
 (ii) Divide and Conquer
 (iii) Greedy approach
 (iv) Back tracking

Code :

- | | |
|---|---|
| A. (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii) | B. (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii) |
| C. (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i) | D. (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i) |

ugcnetcse-july2018-paper2 algorithms

Answer key

1.0.5 UGC NET CSE | January 2017 | Part 3 | Question: 36



Match the following with respect to algorithm paradigms :

List-I

- a. Merge sort
- b. Huffman coding
- c. Optimal polygon triangulation
- d. Subset sum problem

List-II

- i. Dynamic programming
- ii. Greedy approach
- iii. Divide and conquer
- iv. Back tracking

Codes :

- A. a-iii, b-i, c-ii, d-iv
- C. a-ii, b-i, c-iii, d-iv

- B. a-ii, b-i, c-iv, d-iii
- D. a-iii, b-ii, c-i, d-iv

ugcnetcse-jan2017-paper3 algorithms easy match-the-following

Answer key

1.0.6 UGC NET CSE | December 2004 | Part 2 | Question: 25



How much extra space is used by heapsort ?

- A. $O(1)$
- B. $O(\log n)$
- C. $O(n)$
- D. $O(n^2)$

algorithms heap-sort ugcnetcse-dec2004-paper2

Answer key

1.0.7 UGC NET CSE | October 2020 | Part 2 | Question: 90



Given below are two statements:

Statement I: A genetic algorithm is a stochastic hill-climbing search in which a large population of states is maintained

Statement II: In nondeterministic environments, agents can apply AND-OR search to generate contingent plans that reach the goal regardless of which outcomes occur during execution.

In the light of the above statements, choose the correct answer from the options given below

- A. Both Statement I and Statement II are true
- B. Both Statement I and Statement II are false
- C. Statement I is correct but Statement II is false
- D. Statement I is incorrect but Statement II is true

algorithms ugcnetcse-oct2020-paper2

Answer key

1.1 Activity Selection Problem (1)

1.1.1 Activity Selection Problem: UGC NET CSE | December 2015 | Part 3 | Question: 16



In Activity-Selection problem, each activity i has a start time s_i and a finish time f_i where $s_i \leq f_i$. Activities i and j are compatible if

- A. $s_i \geq f_j$
- B. $s_j \geq f_i$
- C. $s_i \geq f_j$ or $s_j \geq f_i$
- D. $s_i \geq f_j$ and $s_j \geq f_i$

ugcnetcse-dec2015-paper3 algorithms activity-selection-problem

Answer key

1.2 Algorithm Design Technique (1)

1.2.1 Algorithm Design Technique: UGC NET CSE | December 2006 | Part 2 | Question: 22



Binary search tree is an example of :

- A. Divide and conquer technique
 C. Back tracking
 B. Greedy algorithm
 D. Dynamic Programming

ugcnetcse-dec2006-paper2 algorithms easy algorithm-design-technique

Answer key 

1.3 Algorithm Efficiency (1)

1.3.1 Algorithm Efficiency: UGC NET CSE | Junet 2015 | Part 2 | Question: 25



To determine the efficiency of an algorithm the time factor is measured by

- A. Counting micro seconds
 C. Counting number of statements
 B. Counting number of key operations
 D. Counting kilobytes of algorithm

ugcnetcse-june2015-paper2 algorithm-efficiency

Answer key 

1.4 Algorithms Sorting (1)

1.4.1 Algorithms Sorting: UGC NET CSE | Junet 2015 | Part 2 | Question: 48



Which of the following algorithms sort n integers, having the range 0 to $(n^2 - 1)$, in ascending order in $O(n)$ time?

- A. Selection sort
 C. Radix sort
 B. Bubble sort
 D. Insertion sort

ugcnetcse-june2015-paper2 algorithms-sorting

Answer key 

1.5 Array (2)

1.5.1 Array: UGC NET CSE | January 2017 | Part 2 | Question: 21



Which of the following is true for computation time in insertion, deletion and finding maximum and minimum element in a sorted array ?

- A. Insertion – $O(1)$, Deletion – $O(1)$, Maximum – $O(1)$, Minimum – $O(1)$
 B. Insertion – $O(1)$, Deletion – $O(1)$, Maximum – $O(n)$, Minimum – $O(n)$
 C. Insertion – $O(n)$, Deletion – $O(n)$, Maximum – $O(1)$, Minimum – $O(1)$
 D. Insertion – $O(n)$, Deletion – $O(n)$, Maximum – $O(n)$, Minimum – $O(n)$

ugcnetjan2017ii algorithms array

Answer key 

1.5.2 Array: UGC NET CSE | July 2016 | Part 3 | Question: 32



Let $A[1, \dots, n]$ be an array of n distinct numbers. If $i < j$ and $A[i] > A[j]$, then the pair (i, j) is called an inversion of A . What is the expected number of inversions in any permutation on n elements?

- A. $\theta(n)$
 B. $\theta(\lg n)$
 C. $\theta(n \lg n)$
 D. $\theta(n^2)$

ugcnetcse-july2016-paper3 algorithms array

Answer key 

1.6 Artificial Intelligence (2)

1.6.1 Artificial Intelligence: UGC NET CSE | June 2014 | Part 3 | Question: 31



Consider $f(N) = g(N) + h(N)$ Where function g is a measure of the cost of getting from the start node to the current node. N and h is an estimate of the additional cost of getting from the current node N to the goal node. Then $f(N) = h(N)$ is used in which one of the following algorithms?

- A. A^* algorithm
C. Greedy best first search algorithm

- B. AO^* algorithm
D. Iterative A^* algorithm

ugcnetjune2014iii artificial-intelligence algorithms

Answer key 

1.6.2 Artificial Intelligence: UGC NET CSE | June 2014 | Part 3 | Question: 33



_____ is used in game trees to reduce the number of branches of the search tree to be traversed without affecting the solution.

- A. Best first search
C. Alpha-beta pruning procedure
- B. Goal stack planning
D. Min-max search

ugcnetjune2014iii artificial-intelligence algorithms

Answer key 

1.7 Asymptotic Notation (8)

1.7.1 Asymptotic Notation: UGC NET CSE | December 2013 | Part 2 | Question: 35



Big-O estimate for $f(x) = (x + 1)\log(x^2 + 1) + 3x^2$ is given as

- A. $O(x\log x)$ B. $O(x^2)$ C. $O(x^3)$ D. $O(x^2\log x)$

ugcnetcse-dec2013-paper2 algorithms easy asymptotic-notation

Answer key 

1.7.2 Asymptotic Notation: UGC NET CSE | December 2013 | Part 3 | Question: 35



Let A and B be two $n \times n$ matrices. The efficient algorithm to multiply the two matrices has the time complexity

- A. $O(n^3)$ B. $O(n^{2.81})$ C. $O(n^{2.67})$ D. $O(n^2)$

ugcnetcse-dec2013-paper3 algorithms asymptotic-notation

Answer key 

1.7.3 Asymptotic Notation: UGC NET CSE | December 2013 | Part 3 | Question: 36



The recurrence relation $T(n) = mT(\frac{n}{2}) + \tan^2$ is satisfied by

- A. $O(n^2)$ B. $O(n^{\lg m})$
C. $O(n^2 \lg n)$ D. $O(n \lg n)$

ugcnetcse-dec2013-paper3 algorithms asymptotic-notation recurrence-relation

Answer key 

1.7.4 Asymptotic Notation: UGC NET CSE | January 2017 | Part 3 | Question: 31



The asymptotic upper bound solution of the recurrence relation given by $T(n) = 2T(\frac{n}{2}) + \frac{n}{\lg n}$ is

- A. $O(n^2)$ B. $O(n \lg n)$
C. $O(n \lg \lg n)$ D. $O(\lg \lg n)$

ugcnetcse-jan2017-paper3 algorithms asymptotic-notation recurrence-relation

Answer key 

1.7.5 Asymptotic Notation: UGC NET CSE | June 2014 | Part 2 | Question: 24



Big-O estimates for the factorial function and the logarithm of the factorial function i.e. $n!$ and $\log n!$ is given by

- A. $O(n!)$ and $O(n\log n)$ B. $O(n^n)$ and $O(n\log n)$
C. $O(n!)$ and $O(\log n!)$ D. $O(n^n)$ and $O(\log n!)$

ugcnetcse-june2014-paper2 algorithms time-complexity asymptotic-notation

Answer key 

1.7.6 Asymptotic Notation: UGC NET CSE | June 2014 | Part 2 | Question: 39



An algorithm is made up of 2 modules M_1 and M_2 . If time complexity of modules M_1 and M_2 are $h(n)$ and $g(n)$ respectively, the time complexity of the algorithm is

- A. $\min(h(n), g(n))$ B. $\max(h(n), g(n))$
C. $h(n) + g(n)$ D. $h(n) * g(n)$

ugcnetcse-june2014-paper2 algorithms time-complexity asymptotic-notation

Answer key 

1.7.7 Asymptotic Notation: UGC NET CSE | June 2014 | Part 3 | Question: 61



Given the following equalities : $E_1 : n^{K+\epsilon} + n^K \lg n = \theta(n^{K+\epsilon})$ for all fixed K and $\epsilon, K \geq 0$ and $\epsilon > 0$.
 $E_2 : n^3 2^n + 6n^2 3^n = O(n^3 2^n)$. Which of the following is true ?

1. E_1 is correct and E_2 is correct.
2. E_1 is correct and E_2 is not correct.
3. E_1 is not correct and E_2 is correct.
4. E_1 is not correct and E_2 is not correct.

asymptotic-notation ugcnetjune2014iii

Answer key 

1.7.8 Asymptotic Notation: UGC NET CSE | Junet 2015 | Part 3 | Question: 35



Let $f(n)$ and $g(n)$ be asymptotically non-negative functions. Which of the following is correct?

- A. $\theta(f(n) * g(n)) = \min(f(n), g(n))$ B. $\theta(f(n) * g(n)) = \max(f(n), g(n))$
C. $\theta(f(n) + g(n)) = \min(f(n), g(n))$ D. $\theta(f(n) + g(n)) = \max(f(n), g(n))$

asymptotic-notation algorithms ugcnetcse-june2015-paper3

Answer key 

1.8 B Tree (2)

1.8.1 B Tree: UGC NET CSE | December 2010 | Part 2 | Question: 21



What is the maximum number of nodes in a B-tree of order 10 of depth 3 (root at depth 0) ?

- A. 111 B. 999 C. 9999 D. None of the above

ugcnetcse-dec2010-paper2 algorithms b-tree

Answer key 

1.8.2 B Tree: UGC NET CSE | December 2013 | Part 2 | Question: 24



For any B-tree of minimum degree $t \geq 2$, every node other than the root must have at least ____ keys and every node can have at most ____ keys.

- A. $t-1, 2t+1$ B. $t+1, 2t+1$ C. $t-1, 2t-1$ D. $t+1, 2t-1$

ugcnetcse-dec2013-paper2 algorithms b-tree

Answer key 

1.9 Bellman Ford (1)

1.9.1 Bellman Ford: UGC NET CSE | October 2020 | Part 2 | Question: 70



Match list I with List II

List I

- (A) Topological sort of DAG
 (B) Kruskal's MST algorithm
 (C) Bellman-Ford's single-source shortest path algorithm
 (D) Floyd-Warshall's all pair shortest path algorithm

List II

- (I) $O(V + E)$
 (II) $O(VE)$
 (III) $\theta(V + E)$
 (IV) $\theta(V^3)$

Choose the correct answer from the options given below:

- A. A-I, B-III, C-IV, D-II
 C. A-III, B-I, C-II, D-IV

- B. A-III, B-I, C-IV, D-II
 D. A-I, B-III, C-II, D-IV

ugcnetcse-oct2020-paper2 topological-sort bellman-ford algorithms match-the-following

Answer key 

1.10

Binary Heap (1)

1.10.1 Binary Heap: UGC NET CSE | December 2012 | Part 3 | Question: 51



Suppose there are $\log_n$ sorted lists of $n \log_n$ element each. The time complexity of producing a sorted list of all these elements is (use heap data structure)

- A. $O(n \log \log_n)$
 C. $\Omega(n \log_n)$
 B. $\theta(n \log_n)$
 D. $\Omega(n^{3/2})$

ugcnetcse-dec2012-paper3 algorithms time-complexity binary-heap

Answer key 

1.11

Binary Search Tree (1)

1.11.1 Binary Search Tree: UGC NET CSE | January 2017 | Part 2 | Question: 25



Which of the following statements is false?

- A. Optimal binary search tree construction can be performed efficiently using dynamic programming.
 B. Breadth-first search cannot be used to find connected components of a graph.
 C. Given the prefix and postfix walks of a binary tree, the tree cannot be reconstructed uniquely.
 D. Depth-first-search can be used to find the connected components of a graph.

ugcnetjan2017ii algorithms binary-tree binary-search-tree

Answer key 

1.12

Binary Tree (4)

1.12.1 Binary Tree: UGC NET CSE | December 2010 | Part 2 | Question: 22



A binary tree with 27 nodes has _____ null branches.

- A. 54
 B. 27
 C. 26
 D. None of the above

ugcnetcse-dec2010-paper2 algorithms binary-tree

Answer key 

1.12.2 Binary Tree: UGC NET CSE | December 2011 | Part 2 | Question: 19



The post order traversal of a binary tree is DEBFCA. Find out the preorder traversal.

A. ABFCDE

B. ADBFEC

C. ABDECF

D. ABDCEF

ugcnetcse-dec2011-paper2 algorithms binary-tree

Answer key **1.12.3 Binary Tree: UGC NET CSE | December 2011 | Part 2 | Question: 50**

The number of nodes in a complete binary tree of height h (with roots at level 0) is equal to

A. $2^0 + 2^1 + \dots + 2^h$

B. $2^0 + 2^1 + \dots + 2^{h-1}$

C. $2^0 + 2^1 + \dots + 2^{h+1}$

D. $2^1 + \dots + 2^{h+1}$

ugcnetcse-dec2011-paper2 algorithms binary-tree

Answer key **1.12.4 Binary Tree: UGC NET CSE | December 2014 | Part 2 | Question: 25**

A full binary tree with n leaves contains

A. n nodesB. $\log_2 n$ nodesC. $2n-1$ nodesD. 2^n nodes

ugcnetcse-dec2014-paper2 algorithms binary-tree

Answer key **1.13****Decision Tree (1)****1.13.1 Decision Tree: UGC NET CSE | December 2014 | Part 3 | Question: 31**

Any decision tree that sorts n elements has height

A. $\Omega(n)$

B. $\Omega(\lg n)$

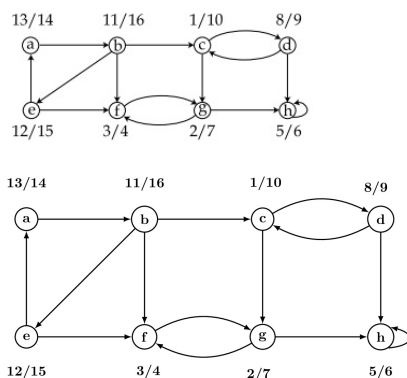
C. $\Omega(n \lg n)$

D. $\Omega(n^2)$

ugcnetcse-dec2014-paper3 algorithms decision-tree

Answer key **1.14****Depth First Search (3)****1.14.1 Depth First Search: UGC NET CSE | December 2015 | Part 2 | Question: 36**

In the following graph, discovery time stamps and finishing time stamps of Depth First Search (DFS) are shown as x/y , where x is discovery time stamp and y is finishing time stamp.



It shows which of the following depth first forest?

A. $\{a,b,e\} \{c,d,f,g,h\}$ B. $\{a,b,e\} \{c,d,h\} \{f,g\}$ C. $\{a,b,e\} \{f,g\} \{c,d\} \{h\}$ D. $\{a,b,c,d\} \{e,f,g\} \{h\}$

ugcnetcse-dec2015-paper2 algorithms depth-first-search

Answer key 

1.14.2 Depth First Search: UGC NET CSE | June 2013 | Part 3 | Question: 68



Which one of the following is not an informed search technique?

- A. Hill climbing search
- B. Best first search
- C. A* search
- D. Depth first search

ugcnetcse-june2013-paper3 algorithms depth-first-search

Answer key

1.14.3 Depth First Search: UGC NET CSE | June 2019 | Part 2 | Question: 65



Which of the following is application of depth-first search?

- A. Only topological sort
- B. Only strongly connected components
- C. Both topological sort and strongly connected components
- D. Neither topological sort nor strongly connected components

ugcnetcse-june2019-paper2 depth-first-search

Answer key

1.15 Dijkstras Algorithm (1)

1.15.1 Dijkstras Algorithm: UGC NET CSE | January 2017 | Part 3 | Question: 35



Dijkstra's algorithm is based on

- A. Divide and conquer paradigm
- B. Dynamic programming
- C. Greedy approach
- D. Backtracking paradigm

ugcnetcse-jan2017-paper3 algorithms dijkstras-algorithm

Answer key

1.16 Double Hashing (1)

1.16.1 Double Hashing: UGC NET CSE | June 2019 | Part 2 | Question: 66



Consider double hashing of the form

$h(k, i) = (h_1(k) + ih_2(k)) \bmod m$ where $h_1(k) = k \bmod m$, $h_2(k) = 1 + (k \bmod n)$ where $n = m - 1$ and $m = 701$. For $k = 123456$, what is the difference between first and second probes in terms of slots?

- A. 255
- B. 256
- C. 257
- D. 258

ugcnetcse-june2019-paper2 double-hashing

Answer key

1.17 Dynamic Programming (2)

1.17.1 Dynamic Programming: UGC NET CSE | December 2013 | Part 3 | Question: 37



The longest common subsequence of the sequences $X = \langle A, B, C, B, D, A, B \rangle$ and $Y = \langle B, D, C, A, B, A \rangle$ has length

- A. 2
- B. 3
- C. 4
- D. 5

ugcnetcse-dec2013-paper3 algorithms dynamic-programming

Answer key

1.17.2 Dynamic Programming: UGC NET CSE | July 2016 | Part 3 | Question: 36



A triangulation of a polygon is a set of T chords that divide the polygon into disjoint triangles. Every triangulation of n vertex convex polygon has ____ chords and divides the polygon into ____ triangles

A. $n-2, n-1$ B. $n-3, n-2$ C. $n-1, n$ D. $n-2, n-2$

ugcnetcse-july2016-paper3 algorithms dynamic-programming

Answer key 

1.18

Graph Algorithms (2)

1.18.1 Graph Algorithms: UGC NET CSE | December 2015 | Part 3 | Question: 20



Floyd-Warshall algorithm utilizes _____ to solve the all-pairs shortest paths problem on a directed graph in _____ time

A. Greedy algorithm, $\theta(V^3)$ B. Greedy algorithm, $\theta(V^2 \lg n)$ C. Dynamic programming, $\theta(V^3)$ D. Dynamic programming, $\theta(V^2 \lg n)$

ugcnetcse-dec2015-paper3 graph-algorithms algorithms

Answer key 

1.18.2 Graph Algorithms: UGC NET CSE | June 2019 | Part 2 | Question: 61



Match List-I with List-II:

	List-I		List-II
(a)	Prim's algorithm	(i)	$O(V^3 \log V)$
(b)	Dijkstra's algorithm	(ii)	$O(VE^2)$
(c)	Faster all-pairs shortest path	(iii)	$O(E \lg V)$
(d)	Edmonds-Karp algorithm	(iv)	$O(V^2)$

Choose the correct option from those options given below:

A. (a) – (ii); (b)-(iv); (c)-(i); (d)-(iii)

B. (a) – (iii); (b)-(iv); (c)-(i); (d)-(ii)

C. (a) – (ii); (b)-(i); (c)-(iv); (d)-(iii)

D. (a) – (iii); (b)-(i); (c)-(iv); (d)-(ii)

ugcnetcse-june2019-paper2 graph-algorithms

Answer key 

1.19

Graph Search (1)

1.19.1 Graph Search: UGC NET CSE | June 2012 | Part 3 | Question: 51



The strategy used to reduce the number of tree branches and the number of static evaluations applied in case of a game tree is

A. Minmax strategy

B. Alpha-beta pruning strategy

C. Constraint satisfaction strategy

D. Static max strategy

ugcnetcse-june2012-paper3 algorithms graph-search tree-traversal

Answer key 

1.20

Greedy Algorithms (1)

1.20.1 Greedy Algorithms: UGC NET CSE | Junet 2015 | Part 3 | Question: 64



Given the symbols A, B, C, D, E, F, G and H with the probabilities $\frac{1}{30}, \frac{1}{30}, \frac{1}{30}, \frac{2}{30}, \frac{3}{30}, \frac{5}{30}, \frac{5}{30}$ and $\frac{12}{30}$ respectively
The average Huffman code size in bits per symbol is

A. $\frac{67}{30}$ B. $\frac{70}{34}$ C. $\frac{76}{30}$ D. $\frac{78}{30}$

ugcnetcse-june2015-paper3 algorithms greedy-algorithms huffman-code

Answer key **1.21****Huffman Code (2)****1.21.1 Huffman Code: UGC NET CSE | July 2016 | Part 3 | Question: 34**

Match the following :

- | | |
|-----------------------------------|--------------------|
| (a) Huffman Code | (i) $O(n^2)$ |
| (b) Optical Polygon Triangulation | (ii) $\theta(n^2)$ |
| (c) Activity Selection Problem | (iii) $O(n \lg n)$ |
| (d) Quicksort | (iv) $\theta(n)$ |

Codes :

- | | |
|---|---|
| A. (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii) | B. (a)-(i), (b)-(iv), (c)-(ii), (d)-(iii) |
| C. (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i) | D. (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i) |

ugcnetcse-july2016-paper3 algorithms huffman-code

Answer key **1.21.2 Huffman Code: UGC NET CSE | July 2016 | Part 3 | Question: 59**

Consider a source with symbols A, B, C, D with probabilities $1/2, 1/4, 1/8, 1/8$ respectively. What is the average number of bits per symbol for the Huffman code generated from above information?

- | | |
|-------------------------|-------------------------|
| A. 2 bits per symbol | B. 1.75 bits per symbol |
| C. 1.50 bits per symbol | D. 1.25 bits per symbol |

ugcnetcse-july2016-paper3 algorithms huffman-code probability

Answer key **1.22****In Place Algo (1)****1.22.1 In Place Algo: UGC NET CSE | June 2019 | Part 2 | Question: 62**

There are many sorting algorithms based on comparison. The running time of heapsort algorithm is $O(n \lg n)$. Like P , but unlike Q , heapsort sorts in place where (P, Q) is equal to

- | | |
|-------------------------------|-------------------------------|
| A. Merge sort, Quick sort | B. Quick sort, insertion sort |
| C. Insertion sort, Quick sort | D. Insertion sort, Merge sort |

ugcnetcse-june2019-paper2 in-place-algo quicksort-mergesort-insertion-sort

Answer key **1.23****Knapsack Problem (2)****1.23.1 Knapsack Problem: UGC NET CSE | June 2014 | Part 3 | Question: 62**

Consider the fractional knapsack instance

$n = 4, (p_1, p_2, p_3, p_4) = (10, 10, 12, 18), (w_1, w_2, w_3, w_4) = (2, 4, 6, 9)$ and $M = 15$.

The maximum profit is given by (Assume p and w denotes profit and weight of objects respectively)

- | | | | |
|-------|-------|-------|-------|
| A. 40 | B. 38 | C. 32 | D. 30 |
|-------|-------|-------|-------|

ugcnetjune2014iii algorithms greedy-algorithm knapsack-problem

Answer key **1.23.2 Knapsack Problem: UGC NET CSE | September 2013 | Part 3 | Question: 42**

Given 0-1 knapsack problem and fractional knapsack problem and the following statements:

S_1 : 0-1 knapsack is efficiently solved using Greedy algorithm.

S_2 : Fractional knapsack is efficiently solved using Dynamic programming.

Which of the following is true?

- A. S_1 is correct and S_2 is not correct
- B. Both S_1 and S_2 are correct
- C. Both S_1 and S_2 are not correct
- D. S_1 is not correct and S_2 is correct

ugcnetcse-sep2013-paper3 algorithms knapsack-problem

Answer key 

1.24

Linear Search (1)

1.24.1 Linear Search: UGC NET CSE | Junet 2015 | Part 2 | Question: 24



The average case occurs in Linear Search Algorithm when

- A. The item to be searched is in some where middle of the Array
- B. The item to be searched is not in the array
- C. The item to be searched is in the last of t he array
- D. The item to be searched is either in the last or not in the array

ugcnetcse-june2015-paper2 algorithms linear-search

Answer key 

1.25

Longest Common Subsequence (1)

1.25.1 Longest Common Subsequence: UGC NET CSE | December 2015 | Part 3 | Question: 17



Given two sequences X and Y :

$X = \langle a, b, c, b, d, a, b \rangle$

$Y = \langle b, d, c, a, b, a \rangle$

The longest common subsequence of X and Y is:

- A. $\langle b, c, a \rangle$
- B. $\langle c, a, b \rangle$
- C. $\langle b, c, a, a \rangle$
- D. $\langle b, c, b, a \rangle$

ugcnetcse-dec2015-paper3 algorithms longest-common-subsequence

Answer key 

1.26

Matrix Chain Ordering (4)

1.26.1 Matrix Chain Ordering: UGC NET CSE | August 2016 | Part 3 | Question: 31



Consider the problem of a chain $\langle A_1, A_2, A_3, A_4 \rangle$ of four matrices. Suppose that the dimensions of the matrices A_1, A_2, A_3 and A_4 are $30 \times 35, 35 \times 15, 15 \times 5$ and 5×10 respectively. The minimum number of scalar multiplications needed to compute the product $A_1 A_2 A_3 A_4$ is ____.

- A. 14875
- B. 21000
- C. 9375
- D. 11875

ugcnetcse-aug2016-paper3 algorithms dynamic-programming numerical-answers matrix-chain-ordering

Answer key 

1.26.2 Matrix Chain Ordering: UGC NET CSE | December 2014 | Part 3 | Question: 35



Consider the problem of a chain $\langle A_1, A_2, A_3 \rangle$ of three matrices. Suppose that the dimensions of the matrices are $10 \times 100, 100 \times 5$ and 5×50 respectively. There are two different ways of parenthesization : (i) $((A_1 A_2) A_3)$ and (ii) $(A_1 (A_2 A_3))$. Computing the product according to the first parenthesization is _____ times faster in comparison to the second parenthesization.

- A. 5
- B. 10
- C. 20
- D. 100

ugcnetcse-dec2014-paper3 algorithms matrix-chain-ordering

Answer key 

1.26.3 Matrix Chain Ordering: UGC NET CSE | January 2017 | Part 3 | Question: 34



The minimum number of scalar multiplication required, for parenthesization of a matrix-chain product whose sequence of dimensions for four matrices is $\langle 5, 10, 3, 12, 5 \rangle$ is

- A. 630 B. 580 C. 480 D. 405

ugcnetcse-jan2017-paper3 algorithms matrix-chain-ordering

Answer key

1.26.4 Matrix Chain Ordering: UGC NET CSE | September 2013 | Part 3 | Question: 39



The number of possible paranthesizations of a sequence of n matrices is

- A. $O(n)$ B. $\theta(n \lg n)$ C. $\Omega(2^n)$ D. None of the above

ugcnetcse-sep2013-paper3 algorithms dynamic-programming matrix-chain-ordering

Answer key

1.27

Merge Sort (2)

1.27.1 Merge Sort: UGC NET CSE | December 2013 | Part 2 | Question: 25



Given two sorted list of size ' m ' and ' n ' respectively. The number of comparisons needed in the worst case by the merge sort algorithm will be

- A. $m \times n$ B. $\max(m, n)$ C. $\min(m, n)$ D. $m+n-1$

ugcnetcse-dec2013-paper2 algorithms merge-sort

Answer key

1.27.2 Merge Sort: UGC NET CSE | June 2014 | Part 2 | Question: 36



Mergesort makes two recursive calls. Which statement is true after these two recursive calls finish, but before the merge step ?

- A. The array elements form a heap.
B. Elements in each half of the array are sorted amongst themselves.
C. Elements in the first half of the array are less than or equal to elements in second half of the array.
D. All of the above

ugcnetcse-june2014-paper2 algorithms sorting merge-sort

Answer key

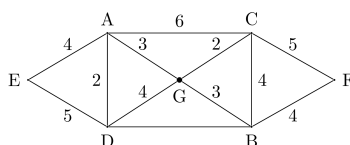
1.28

Minimum Spanning Tree (1)

1.28.1 Minimum Spanning Tree: UGC NET CSE | November 2017 | Part 2 | Question: 5



Consider the graph given below:



Use Kruskal's algorithm to find a minimal spanning tree for the graph. The List of the edges of the tree in the order in which they are chosen is

- A. AD, AE, AG, GC, GB, BF B. GC, GB, BF, GA, AD, AE
C. GC, AD, GB, GA, BF, AE D. AD, AG, GC, AE, GB, BF

ugcnetcse-nov2017-paper2 graph-algorithms minimum-spanning-tree

1.29.1 Non Gatecse: UGC NET CSE | October 2020 | Part 2 | Question: 69



Match List I with List II

With reference to CMM developed by Software Engineering Institute (SEI)

List I

(A) Branch-and-bound

(B) Steepest-descent hill climbing

(C) Constraint satisfaction

(D) Means-end-analysis

List II

(I) Keeps track of all partial paths which can be candidate for further exploration.

(II) Detects difference between current state and goal state

(III) Discovers problem state(s) that satisfy a set of constraints

(IV) Considers all moves from current state and selects best move

Choose the correct answer from the options given below:

A. A-I, B-IV, C-III, D-II

C. A-II, B-I, C-III, D-IV

B. A-I, B-II, C-III, D-IV

D. A-II, B-IV, C-III, D-I

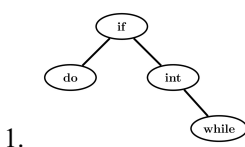
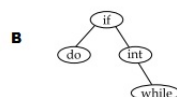
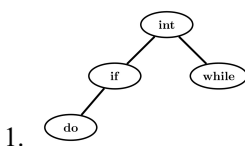
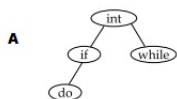
ugcnetcse-oct2020-paper2 non-gatecse algorithms

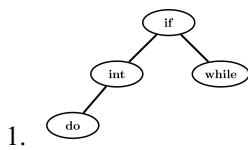
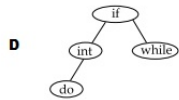
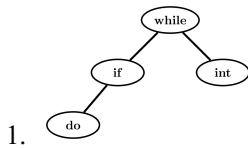
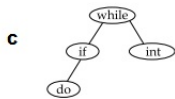
Answer key 

1.30.1 Optimal Search Tree: UGC NET CSE | December 2015 | Part 3 | Question: 21



Let $n = 4$ and $(a_1, a_2, a_3, a_4) = (\text{do}, \text{if}, \text{int}, \text{while})$. Let $p(1:4) = \left(\frac{3}{8}, \frac{3}{8}, \frac{1}{8}, \frac{1}{8}\right)$ and $q(1:4) = \left(\frac{2}{8}, \frac{3}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}\right)$ where $p(i)$ and $q(i)$ denotes the probability with which we search a_i and the identifier x being searched satisfy $a_i < x < a_{i+1}$ respectively. The optimal search tree is given by:





ugcnetcse-dec2015-paper3 algorithms optimal-search-tree

Answer key

1.31

Optimal Solution (1)

1.31.1 Optimal Solution: UGC NET CSE | June 2019 | Part 2 | Question: 68



Consider the following steps:

S_1 : Characterize the structure of an optimal solution

S_2 : Compute the value of an optimal solution in bottom-up fashion

Which of the following step(s) is/are common to both dynamic programming and greedy algorithms?

- A. Only S_1 B. Only S_2 C. Both S_1 and S_2 D. Neither S_1 nor S_2

ugcnetcse-june2019-paper2 optimal-solution dynamic-programming greedy-algorithm

Answer key

1.32

P NP NPC NPH (3)

1.32.1 P NP NPC NPH: UGC NET CSE | December 2014 | Part 3 | Question: 33



We can show that the clique problem is NP -hard by proving that

- A. $CLIQUE \leq P$ 3-CNF_SAT B. $CLIQUE \leq P$ VERTEX_COVER
C. $CLIQUE \leq P$ SUBSET_SUM D. None of the above

ugcnetcse-dec2014-paper3 algorithms p-np-npc-nph

Answer key

1.32.2 P NP NPC NPH: UGC NET CSE | June 2019 | Part 2 | Question: 67



Consider the complexity class $CO - NP$ as the set of languages L such that $\bar{L} \in NP$, and the following two statements:

S_1 : $P \subseteq CO - NP$

S_2 : If $NP \neq CO - NP$, then $P \neq NP$

Which of the following is/are correct?

- A. Only S_1 B. Only S_2 C. Both S_1 and S_2 D. Neither S_1 nor S_2

Answer key

1.32.3 P NP NPC NPH: UGC NET CSE | Junet 2015 | Part 3 | Question: 32

The travelling salesman problem can be solved in

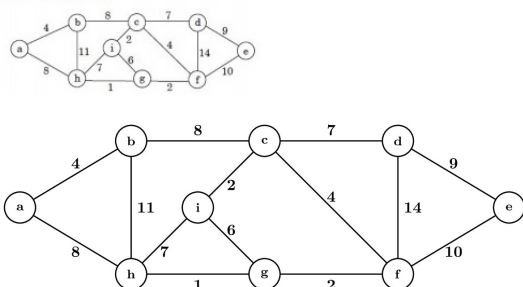
- A. Polynomial time using dynamic programming algorithm
- B. Polynomial time using branch-and-bound algorithm
- C. Exponential time using dynamic programming algorithm or branch-and-bound algorithm
- D. Polynomial time using back tracking algorithm

algorithms p-np-npc-nph ugcnetcse-june2015-paper3

Answer key

1.33**Prims Algorithm (1)****1.33.1 Prims Algorithm: UGC NET CSE | October 2020 | Part 2 | Question: 78**

Consider the undirected graph below:



Using Prim's algorithm to construct a minimum spanning tree starting with node a , which one of the following sequences of edges represents a possible order in which the edges would be added to construct the minimum spanning tree?

- A. $(a, b), (a, h), (g, h), (f, g), (c, f), (c, i), (c, d), (d, e)$
- B. $(a, b), (b, h), (g, h), (g, i), (c, i), (c, f), (c, d), (d, e)$
- C. $(a, b), (b, c), (c, i), (c, f), (f, g), (g, h), (c, d), (d, e)$
- D. $(a, b), (g, h), (g, f), (c, f), (c, i), (f, e), (b, c), (d, e)$

prims-algorithm algorithms minimum-spanning-tree ugcnetcse-oct2020-paper2

Answer key

1.34**Quick Sort (1)****1.34.1 Quick Sort: UGC NET CSE | June 2014 | Part 3 | Question: 66**

Suppose that the splits at every level of quicksort are in the proportion $(1-\alpha)$ to α , where $0 < \alpha \leq \frac{1}{2}$ is a constant. The minimum depth of a leaf in the recursion tree is approximately given by

- A. $-\frac{\lg n}{\lg(1-\alpha)}$
- B. $-\frac{\lg(1-\alpha)}{\lg n}$
- C. $-\frac{\lg n}{\lg \alpha}$
- D. $-\frac{\lg \alpha}{\lg n}$

ugcnetjune2014iii data-structures sorting algorithms quick-sort

Answer key

1.35**Recurrence Relation (7)**

B. $T(n) = O(\log_2 n)$
D. $T(n) = O(n^2)$

Answer key 

$$T(n) \leq \begin{cases} \theta(1) & \text{if } n \leq 80 \\ T\left(\frac{n}{s}\right) + T\left(\frac{7n}{10} + 6\right) + O(n) & \text{if } n > 80 \end{cases} \text{ is}$$

- Answer key

B. $O(n \lg(n))$
D. $O(\lg(n) \lg(\lg(n)))$

Answer key

Answer key 

B. $O(n \log \log n)$
D. $O(\log n \log \log n)$

Answer key

Answer key 

1.35.7 Recurrence Relation: UGC NET CSE | September 2013 | Part 3 | Question: 37



The time complexity of recurrence relation $T(n) = T(n/3) + T(2n/3) + O(n)$ is

- A. $O(\lg n)$ B. $O(n)$ C. $O(n \lg n)$ D. $O(n^2)$

ugcnetcse-sep2013-paper3 algorithms time-complexity recurrence-relation

Answer key

1.36 Recursion (2)

1.36.1 Recursion: UGC NET CSE | December 2009 | Part 2 | Question: 11



Recursive functions are executed in a

- (A) First in first out-order
(B) Last in first out-order
(C) Parallel fashion
(D) Load balancing

ugcnetcse-dec2009-paper2 algorithms recursion

Answer key

1.36.2 Recursion: UGC NET CSE | December 2011 | Part 2 | Question: 3



Which of the following is a bad example of recursion ?

- A. Factorial B. Fibonacci numbers
C. Tower of Hanai D. Tree traversal

ugcnetcse-dec2011-paper2 algorithms recursion

Answer key

1.37 Red Black Tree (1)

1.37.1 Red Black Tree: UGC NET CSE | January 2017 | Part 3 | Question: 33



Red-black trees are one of many search tree schemes that are “balanced” in order to guarantee that basic dynamic-set operations take _____ time in the worst case.

- A. $O(1)$ B. $O(\lg n)$ C. $O(n)$ D. $O(n \lg n)$

ugcnetcse-jan2017-paper3 algorithms red-black-tree

Answer key

1.38 Shortest Path (2)

1.38.1 Shortest Path: UGC NET CSE | December 2014 | Part 3 | Question: 34



Dijkstra algorithm, which solves the single-source shortest-paths problem, is a _____, and the Floyd-Warshall algorithm, which finds shortest paths between all pairs of vertices, is a _____.

- A. Greedy algorithm, Divide-conquer algorithm
B. Divide-conquer algorithm, Greedy algorithm
C. Greedy algorithm, Dynamic programming algorithm
D. Dynamic programming algorithm, Greedy algorithm

ugcnetcse-dec2014-paper3 algorithms shortest-path

Answer key

1.38.2 Shortest Path: UGC NET CSE | Junet 2015 | Part 3 | Question: 31



An all-pairs shortest-paths problem is efficiently solved using:

- A. Dijkstra's algorithm
- B. Bellman-Ford algorithm
- C. Kruskal algorithm
- D. Floyd-Warshall algorithm

ugcnetcse-june2015-paper3 algorithms easy shortest-path

Answer key

1.39 Sorting (4)

1.39.1 Sorting: UGC NET CSE | December 2013 | Part 3 | Question: 38



Assuming there are n keys and each key is in the range $[0, m-1]$. The run time of bucket sort is

- A. $O(n)$
- B. $O(n \lg n)$
- C. $O(n \lg m)$
- D. $O(n+m)$

ugcnetcse-dec2013-paper3 algorithms sorting

Answer key

1.39.2 Sorting: UGC NET CSE | December 2014 | Part 2 | Question: 22



You have to sort a list L , consisting of a sorted list followed by a few 'random' elements. Which of the following sorting method would be most suitable for such a task ?

- A. Bubble sort
- B. Selection sort
- C. Quick sort
- D. Insertion sort

ugcnetcse-dec2014-paper2 algorithms data-structures sorting

Answer key

1.39.3 Sorting: UGC NET CSE | December 2015 | Part 3 | Question: 18



If there are n integers to sort, each integer had d digits and each digit is in the set $\{1, 2, \dots, k\}$, radix sort can sort the numbers in

- A. $O(dnk)$
- B. $O(dn^k)$
- C. $O(d+n)k$
- D. $O(d(n+k))$

ugcnetcse-dec2015-paper3 algorithms sorting

Answer key

1.39.4 Sorting: UGC NET CSE | June 2019 | Part 2 | Question: 64



Which of the following is best running time to sort n integers in the range 0 to $n^2 - 1$?

- A. $O(\lg n)$
- B. $O(n)$
- C. $O(n \lg n)$
- D. $O(n^2)$

ugcnetcse-june2019-paper2 sorting

Answer key

1.40 Space Complexity (2)

1.40.1 Space Complexity: UGC NET CSE | December 2015 | Part 2 | Question: 39



An ideal sort is an in-place-sort whose additional space requirement is

- A. $O(\log_2 n)$
- B. $O(n \log_2 n)$
- C. $O(1)$
- D. $O(n)$

ugcnetcse-dec2015-paper2 algorithms sorting space-complexity

Answer key

1.40.2 Space Complexity: UGC NET CSE | October 2020 | Part 2 | Question: 73



Match List I with List II

List I

(A) Greedy Best-First Search

(B) A^*

(C) Recursive Best-First Search

(D) SMA*

List II

(I) Space complexity is $O(d)$ where d =depth of the deepest optimal solution

(II) Incomplete even if the search space is finite

(III) Optimal if optimal solution is reachable; otherwise, returns the best reachable optimal solution

(IV) Computation and space complexity is too high

Choose the correct answer from the options given below:

A. A-II, B-IV, C-I, D-III

C. A-III, B-II, C-IV, D-I

B. A-II, B-III, C-I, D-IV

D. A-III, B-IV, C-II, D-I

ugcnetcse-oct2020-paper2 space-complexity algorithms

Answer key

1.41

Time Complexity (10)

1.41.1 Time Complexity: UGC NET CSE | December 2006 | Part 2 | Question: 21



Which algorithm has same average, worst case and best case time ?

A. Binary search

C. Quick sort

B. Maximum of n number

D. Fibonacci search

algorithms time-complexity ugcnetcse-dec2006-paper2

Answer key

1.41.2 Time Complexity: UGC NET CSE | December 2010 | Part 2 | Question: 23



The time complexity to build a heap of n elements is

A. $O(1)$

B. $O(\lg n)$

C. $O(n)$

D. $O(n \lg n)$

ugcnetcse-dec2010-paper2 algorithms time-complexity

Answer key

1.41.3 Time Complexity: UGC NET CSE | December 2012 | Part 3 | Question: 11



The time complexities of some standard graph algorithms are given. Match each algorithm with its time complexity ? (n and m are no. of nodes and edges respectively)

a. Bellman Ford algorithm

b. Kruskals algorithm

c. Floyd Warshall algorithm

d. Topological sorting

i. $O(m \log n)$

ii. $O(n^3)$

iii. $O(mn)$

iv. $O(n + m)$

Codes :

A. a-iii, b-i, c-ii, d-iv

C. a-iii, b-iv, c-i, d-ii

B. a-ii, b-iv, c-iii, d-i

D. a-ii, b-i, c-iii, d-iv

ugcnetcse-dec2012-paper3 algorithms time-complexity

Answer key

1.41.4 Time Complexity: UGC NET CSE | December 2014 | Part 3 | Question: 32



Match the following :

List – I

- a. Bucket sort
- b. Matrix chain multiplication
- c. Huffman codes
- d. All pairs shortest paths

List – II

- i. $O(n^3 \lg n)$
- ii. $O(n^3)$
- iii. $O(n \lg n)$
- iv. $O(n)$

Codes :

- A. a-iv, b-ii, c-i, d-iii
- C. a-iv, b-ii, c-iii, d-i

- B. a-ii, b-iv, c-i, d-iii
- D. a-iii, b-ii, c-iv, d-i

ugcnetcse-dec2014-paper3 algorithms time-complexity

Answer key

1.41.5 Time Complexity: UGC NET CSE | Junet 2015 | Part 3 | Question: 33



Which of the following is asymptotically smaller?

- A. $\lg(\lg^* n)$
- B. $\lg^*(\lg n)$
- C. $\lg(n!)$
- D. $\lg^*(n!)$

algorithms ugcnetcse-june2015-paper3 time-complexity

Answer key

1.41.6 Time Complexity: UGC NET CSE | October 2020 | Part 2 | Question: 25



If algorithm A and another algorithm B take $\log_2(n)$ and $\sqrt{n}$ microseconds, respectively, to solve a problem, then the largest size n of a problem these algorithms can solve, respectively, in one second are _____ and _____.

- A. 2^{10^n} and 10^6
- B. 2^{10^n} and 10^{12}
- C. 2^{10^n} and 6.10^6
- D. 2^{10^n} and 6.10^{12}

ugcnetcse-oct2020-paper2 algorithms time-complexity

Answer key

1.41.7 Time Complexity: UGC NET CSE | October 2020 | Part 2 | Question: 52



The running time of an algorithm is $O(g(n))$ if and only if

- a. its worst-case running time is $O(g(n))$ and its best-case running time is $\Omega(g(n))$ ($O = \text{big } O$)
- b. its worst-case running time is $\Omega(g(n))$ and its best-case running time is $O(g(n))$ ($O = \text{big } O$)
- c. $O(g(n)) = \Omega(g(n))$ ($O = \text{big } O$)
- d. $O(g(n)) \cap \omega(g(n))$ is non-empty set, ($o = \text{small } o$)

ugcnetcse-oct2020-paper2 algorithms time-complexity

Answer key

1.41.8 Time Complexity: UGC NET CSE | October 2022 | Part 1 | Question: 87



Consider the following algorithms and their running times :

Algorithms	Complexities
(A) Breadth First Search	(I) $\theta(v + E)$
(B) Rabin-Karp Algorithm	(II) $O(v + E)$
(C) Depth-First Search	($\begin{smallmatrix} \text{I} & \text{I} & \text{I} \\ \theta((n - m - 1)m) \end{smallmatrix}$)

(D) Heap sort (worst case)	(IV) $O(n^2)$
(E) Quick sort (worst case)	(V) $O(n \lg n)$

Which one of the following is correct?

- A. (A)-(III), (B)-(II), (C)-(I), (D)-(IV), (E)-(V)
- B. (A)-(II), (B)-(III), (C)-(I), (D)-(IV), (E)-(V)
- C. (A)-(II), (B)-(III), (C)-(I), (D)-(V), (E)-(IV)
- D. (A)-(III), (B)-(I), (C)-(II), (D)-(IV), (E)-(V)

ugcnetcse-oct2022-paper1 algorithms time-complexity match-the-following

Answer key 

1.41.9 Time Complexity: UGC NET CSE | September 2013 | Part 2 | Question: 9



The amortized time complexity to perform ____ operation(s) in Splay trees is $O(\lg n)$

- A. Search
- B. Search and Insert
- C. Search and Delete
- D. Search, Insert and Delete

ugcnetsep2013iii algorithms time-complexity

Answer key 

1.41.10 Time Complexity: UGC NET CSE | September 2013 | Part 3 | Question: 40



The time complexity of an efficient algorithm to find the longest monotonically increasing subsequence of n numbers is

- A. $O(n)$
- B. $O(n \lg n)$
- C. $O(n^2)$
- D. None of the above

algorithms time-complexity ugcnetcse-sep2013-paper3

Answer key 

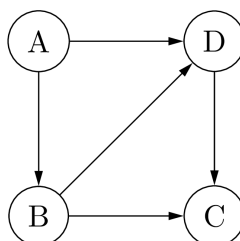
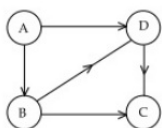
1.42

Topological Sort (1)

1.42.1 Topological Sort: UGC NET CSE | December 2005 | Part 2 | Question: 23



Consider the graph, which of the following is a valid topological sorting?



- A. ABCD
- B. BACD
- C. BADC
- D. ABDC

ugcnetcse-dec2005-paper2 algorithms topological-sort

Answer key 

1.43

Tree (2)

1.43.1 Tree: UGC NET CSE | December 2015 | Part 2 | Question: 37



The number of disk pages access in B-tree search, where h is height, n is the number of keys and t is the minimum degree, is

- A. $\theta(\log_n h * t)$ B. $\theta(\log_t n * h)$
 C. $\theta(\log_h n)$ D. $\theta(\log_t n)$

ugcnetcse-dec2015-paper2 algorithms tree

Answer key

1.43.2 Tree: UGC NET CSE | January 2017 | Part 3 | Question: 32



Any decision tree that sorts n elements has height _____

- A. $\Omega(\lg n)$ B. $\Omega(n)$
 C. $\Omega(n \lg n)$ D. $\Omega(n^2)$

ugcnetcse-jan2017-paper3 algorithms tree

Answer key

1.44

Tree Search Algorithm (2)

1.44.1 Tree Search Algorithm: UGC NET CSE | Junet 2015 | Part 3 | Question: 58



Which of the following statements is true for Branch-and-Bound search?

- A. Underestimates of remaining distance may cause deviation from optimal path
 B. Overestimates can't cause right path to be overlooked
 C. Dynamic programming principle can be used to discard redundant partial paths
 D. All of the above

ugcnetcse-june2015-paper3 algorithms branch-and-bound tree-search-algorithm graph-search

Answer key

1.44.2 Tree Search Algorithm: UGC NET CSE | September 2013 | Part 3 | Question: 2



$\alpha - \beta$ cutoffs are applied to

- A. Depth first search B. Best first search
 C. Minimax search D. Breadth first search

ugcnetcse-sep2013-paper3 algorithms graph-search tree-search-algorithm

Answer key

1.45

Tree Traversal (1)

1.45.1 Tree Traversal: UGC NET CSE | Junet 2015 | Part 2 | Question: 4



Consider the following statements:

- i. Depth-first search is used to traverse a rooted tree
- ii. Pre-order, Post-order and Inorder are used to list the vertices of an ordered rooted tree.
- iii. Huffman's algorithm is used to find an optimal binary tree with given weights
- iv. Topological sorting provides a labelling such that the parents have larger labels than their children

Which one of the above statements is true?

- A. i and ii B. iii and iv
 C. i, ii, and iii D. i, ii, iii, and iv

ugcnetcse-june2015-paper2 tree tree-traversal

Answer key

Answer Keys

1.0.1	C	1.0.2	Q-Q	1.0.3	Q-Q	1.0.4	Q-Q	1.0.5	D
1.0.6	A	1.0.7	A	1.1.1	C	1.2.1	Q-Q	1.3.1	B
1.4.1	C	1.5.1	C	1.5.2	D	1.6.1	C	1.6.2	C
1.7.1	B	1.7.2	B	1.7.3	X	1.7.4	C	1.7.5	B
1.7.6	B	1.7.7	B	1.7.8	D	1.8.1	Q-Q	1.8.2	C
1.9.1	X	1.10.1	A	1.11.1	B	1.12.1	Q-Q	1.12.2	Q-Q
1.12.3	Q-Q	1.12.4	C	1.13.1	C	1.14.1	A	1.14.2	D
1.14.3	C	1.15.1	C	1.16.1	C	1.17.1	C	1.17.2	B
1.18.1	A	1.18.2	B	1.19.1	B	1.20.1	C	1.21.1	C
1.21.2	B	1.22.1	D	1.23.1	B	1.23.2	C	1.24.1	A
1.25.1	B	1.26.1	Q-Q	1.26.2	B	1.26.3	D	1.26.4	C
1.27.1	D	1.27.2	B	1.28.1	C	1.29.1	A	1.30.1	C
1.31.1	A	1.32.1	D	1.32.2	C	1.32.3	C	1.33.1	A;C
1.34.1	C	1.35.1	C	1.35.2	C	1.35.3	D	1.35.4	Q-Q
1.35.5	D	1.35.6	C	1.35.7	C	1.36.1	Q-Q	1.36.2	B
1.37.1	B	1.38.1	C	1.38.2	D	1.39.1	D	1.39.2	D
1.39.3	D	1.39.4	B	1.40.1	C	1.40.2	A	1.41.1	Q-Q
1.41.2	Q-Q	1.41.3	A	1.41.4	C	1.41.5	A	1.41.6	B
1.41.7	X	1.41.8	N/A	1.41.9	D	1.41.10	B	1.42.1	Q-Q
1.43.1	D	1.43.2	C	1.44.1	C	1.44.2	C	1.45.1	D